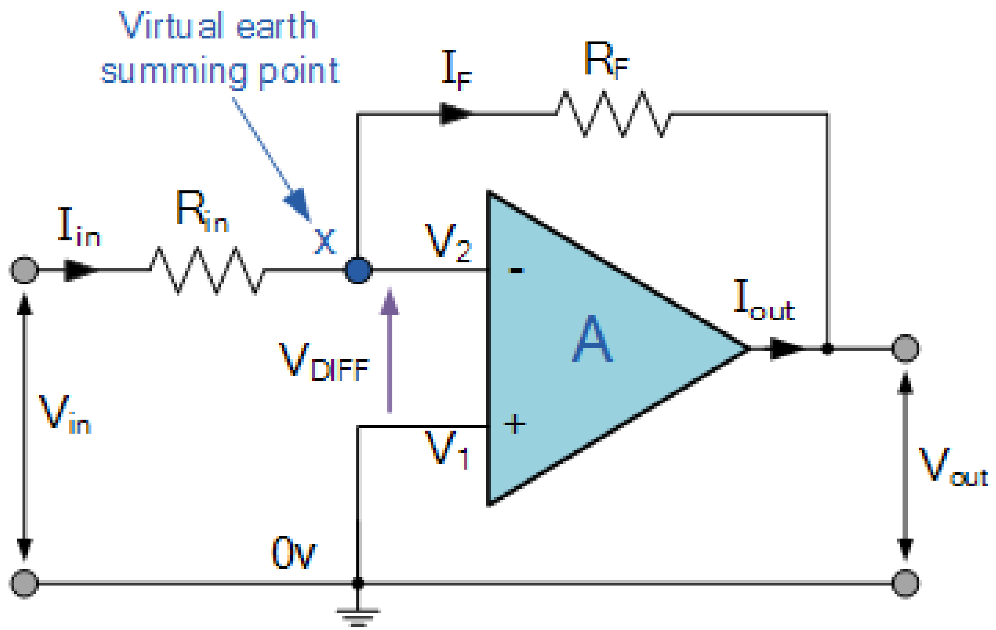
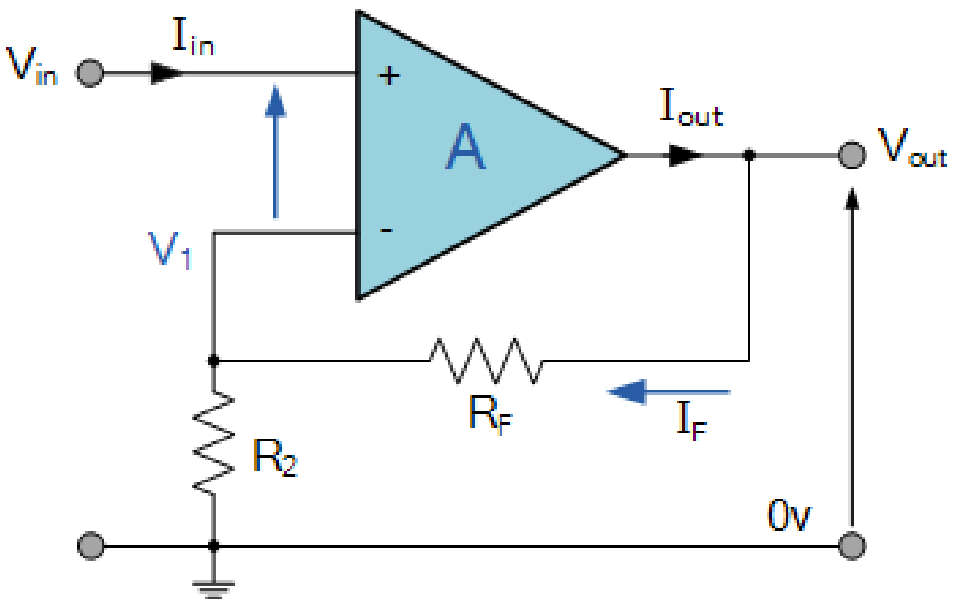
Circuits for Sensors

Inverting Operational Amplifier



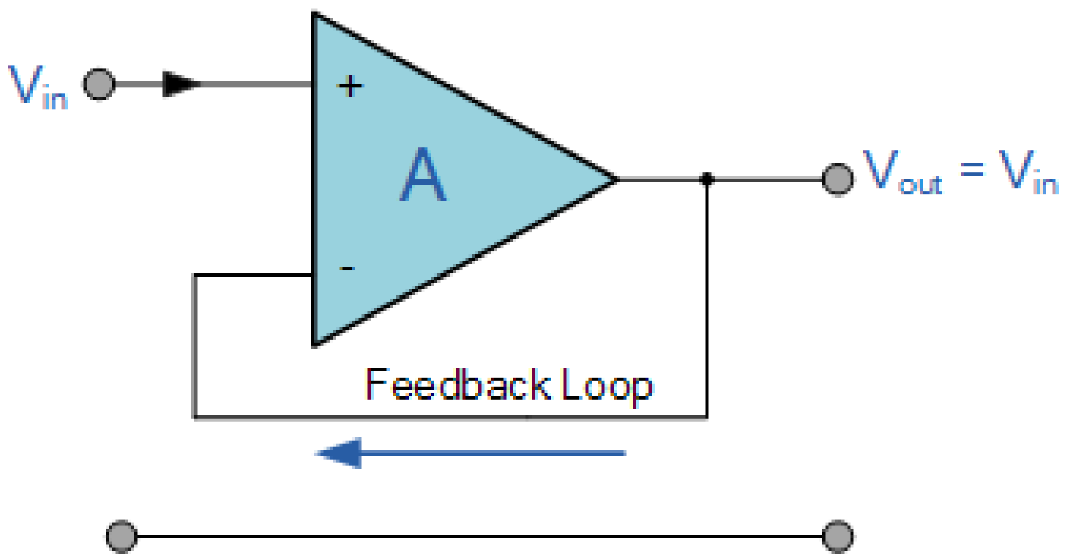
$$A_v = \frac{V_{out}}{V_{in}} = -\frac{R_f}{R_{in}}$$

Non-inverting Operational Amplifier



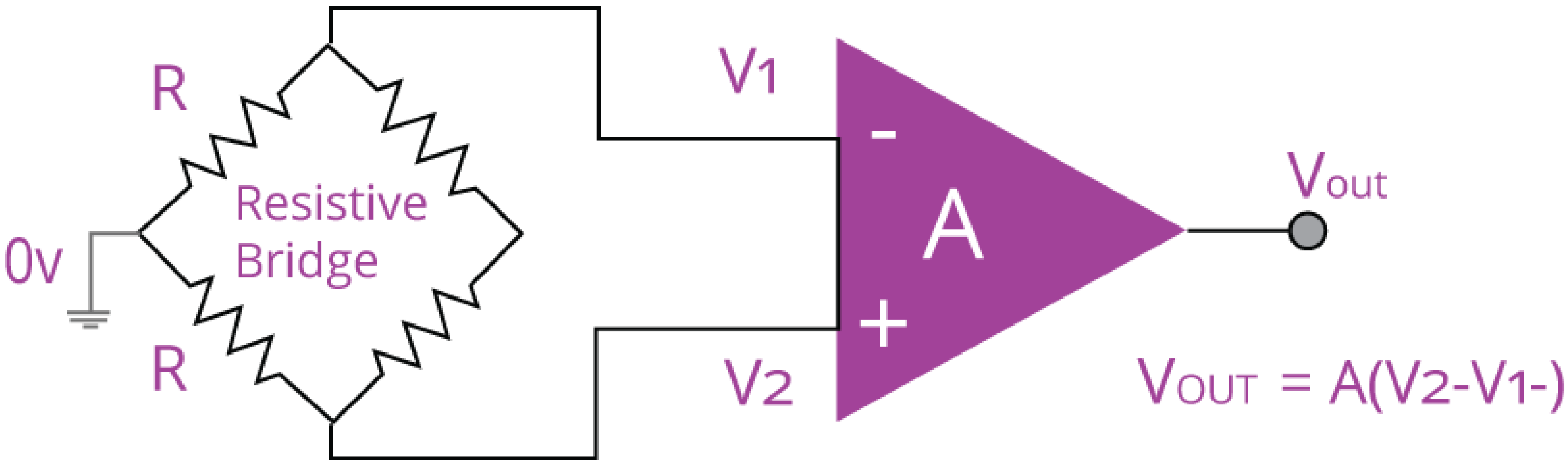
$$A_v = 1 + \frac{R_F}{R_2}$$

Voltage follower (pre-amplifier)



$$A_v = 1$$

Differential Amplifier



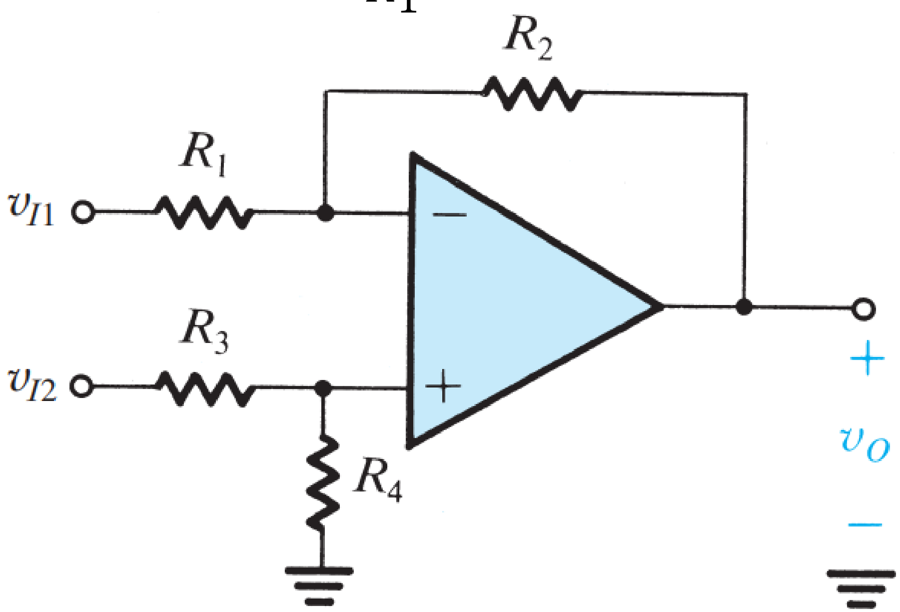
Differential Amplifier

- Op-Amp as Differential Amplifier

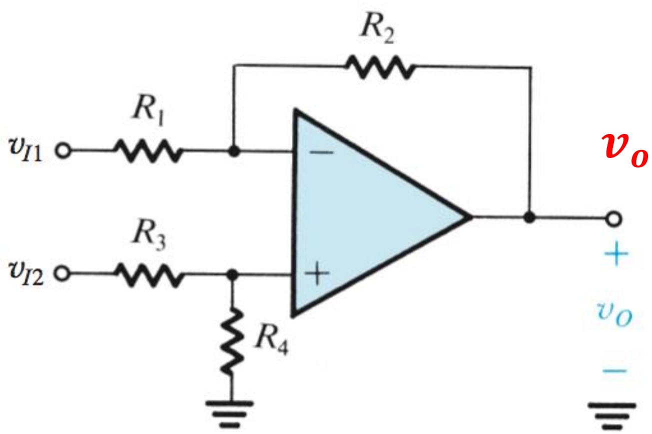
$$v_0 = \frac{R_4}{R_3 + R_4}\left(\frac{R_1 + R_2}{R_1}\right)v_{i2} - \frac{R_2}{R_1}v_{i1}$$

If $R_1 = R_3$ and $R_2 = R_4$

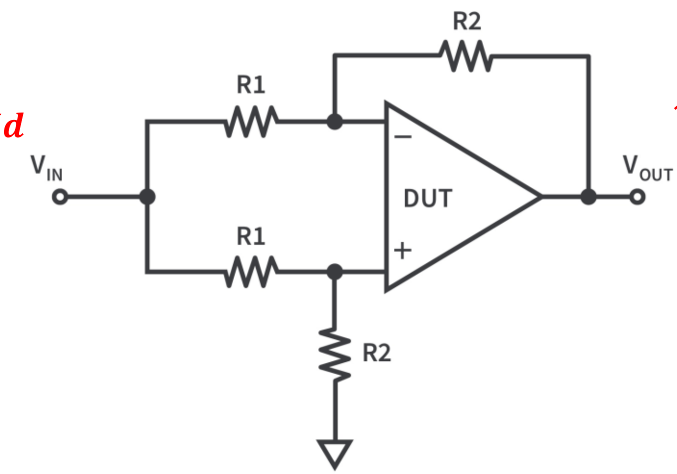
$$v_0 = \frac{R_2}{R_1}(v_{i2} - v_{i1})$$



Differential Amplifier



$v_o = A_d v_{Id}$



$v_o = A_{cm} v_{Icm}$

Differential gain

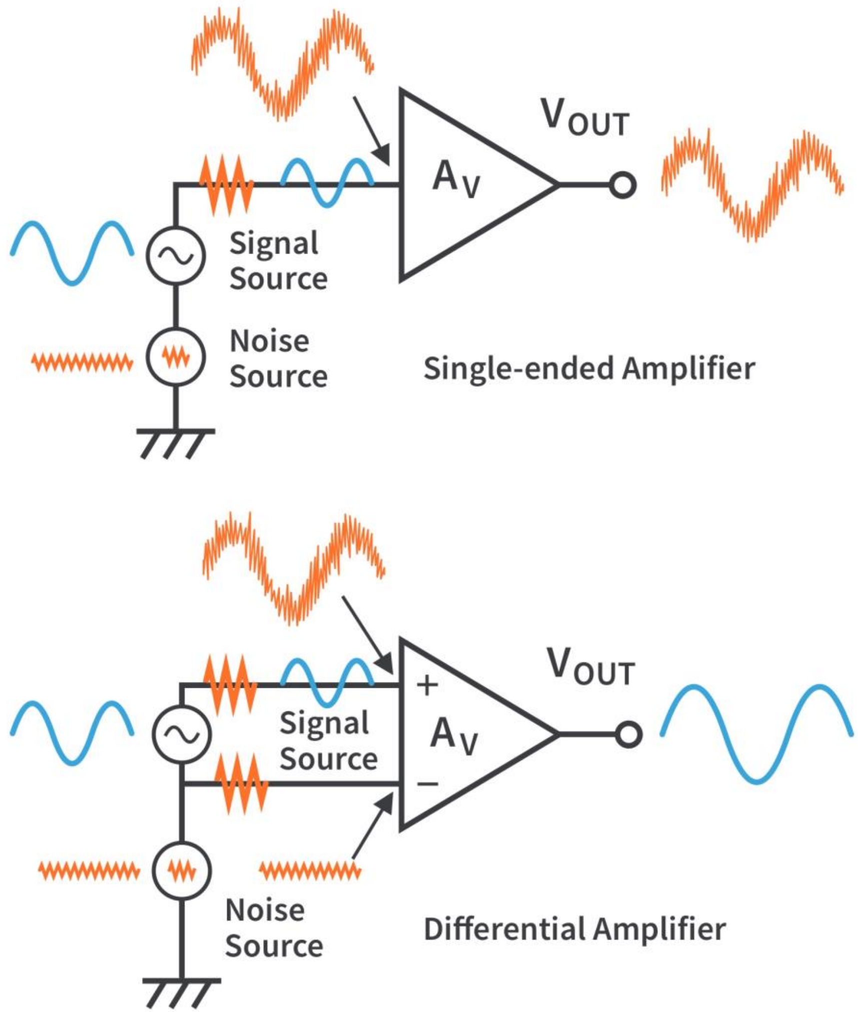
Common-mode gain

$$v_o = A_d v_{Id} + A_{cm} v_{Icm}$$

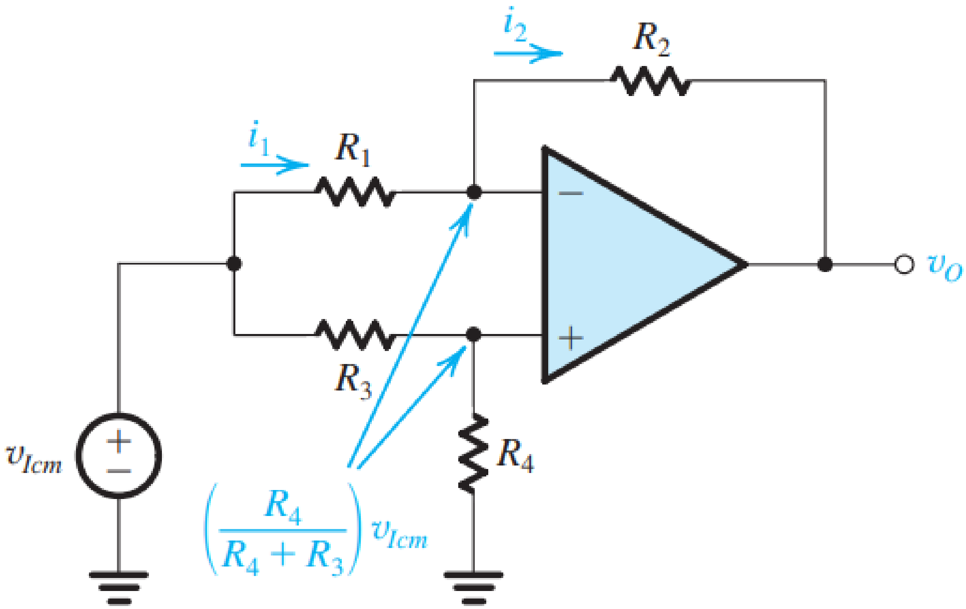
Common-Mode Rejection Ratio (CMRR)

$$CMRR = 20\log\frac{A_d}{A_{cm}}$$

Differential Amplifier



Differential Amplifier



$$v_o = \frac{R_4}{R_4+R_3}\left(1-\frac{R_2R_3}{R_1R_4}\right)v_{Icm}$$

Common mode gain

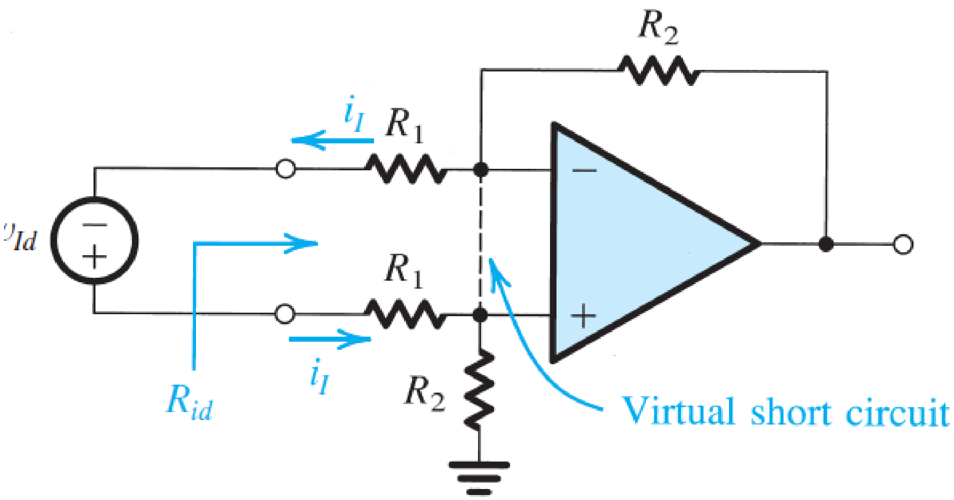
$$A_{cm} = \frac{v_o}{v_{Icm}} = \left(\frac{R_4}{R_4+R_3}\right)\left(1-\frac{R_2}{R_1}\frac{R_3}{R_4}\right)$$

Ideally, $R_1 = R_3$ and $R_2 = R_4$

$$A_{cm} = 0 \text{ and } CMRR = \infty$$

Any mismatch in the resistance ratios can make A_{cm} nonzero

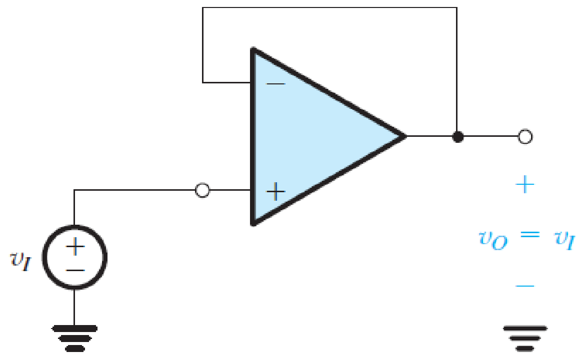
Differential Amplifier

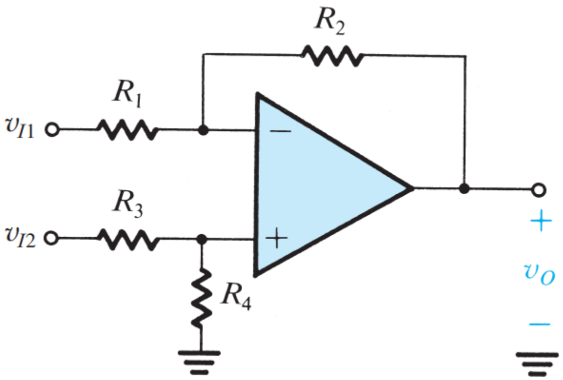


- **Differential input resistance**

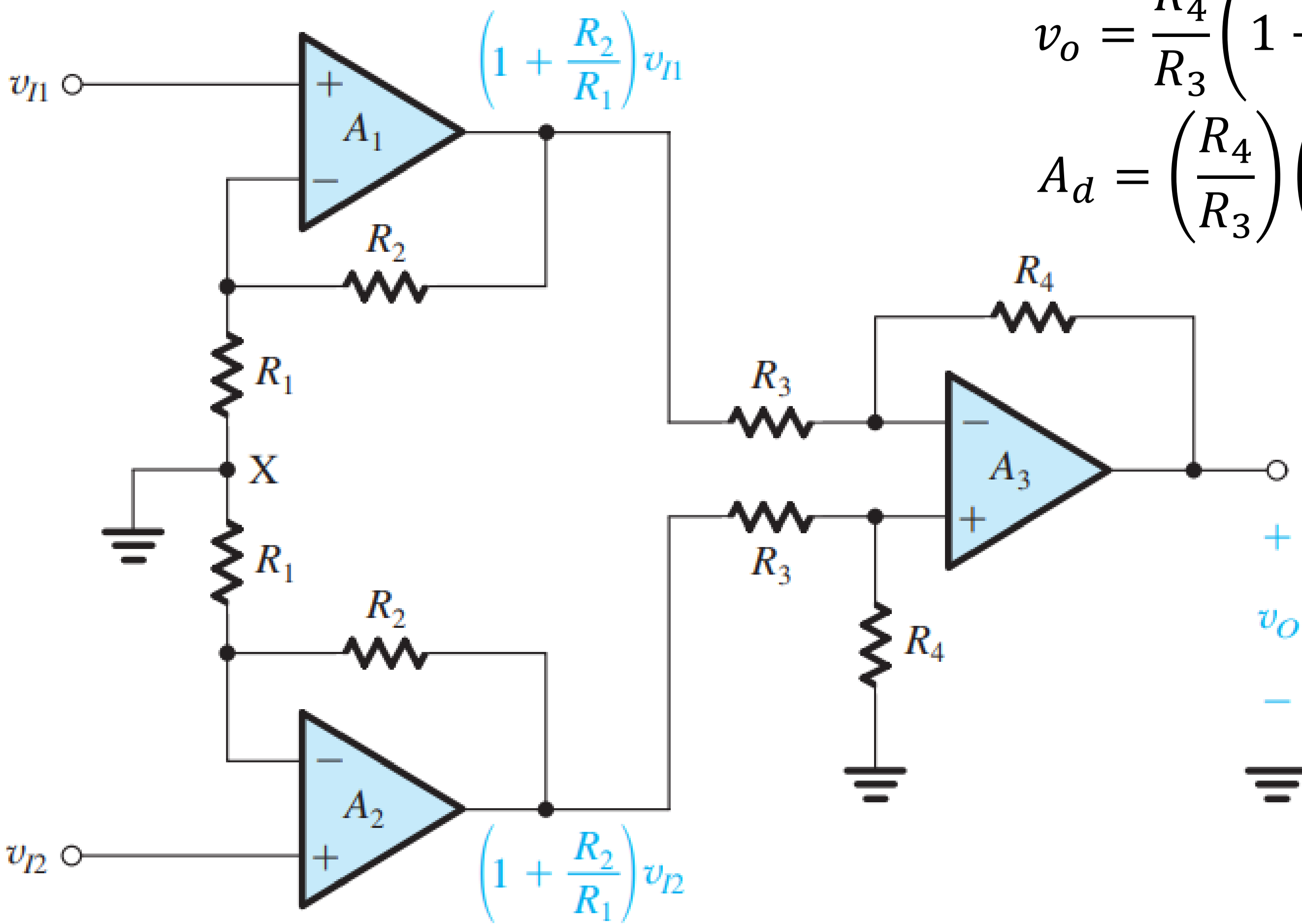
$$R_{id} \equiv \frac{v_{Id}}{i_I}$$
$$v_{Id} = R_1 i_I + R_1 i_I$$
$$R_{id} = 2R_1$$

- High input impedance: ✕
- Low output impedance: ✔
- High gain: ✕
- High CMRR: ✕





Instrumentation Amplifier



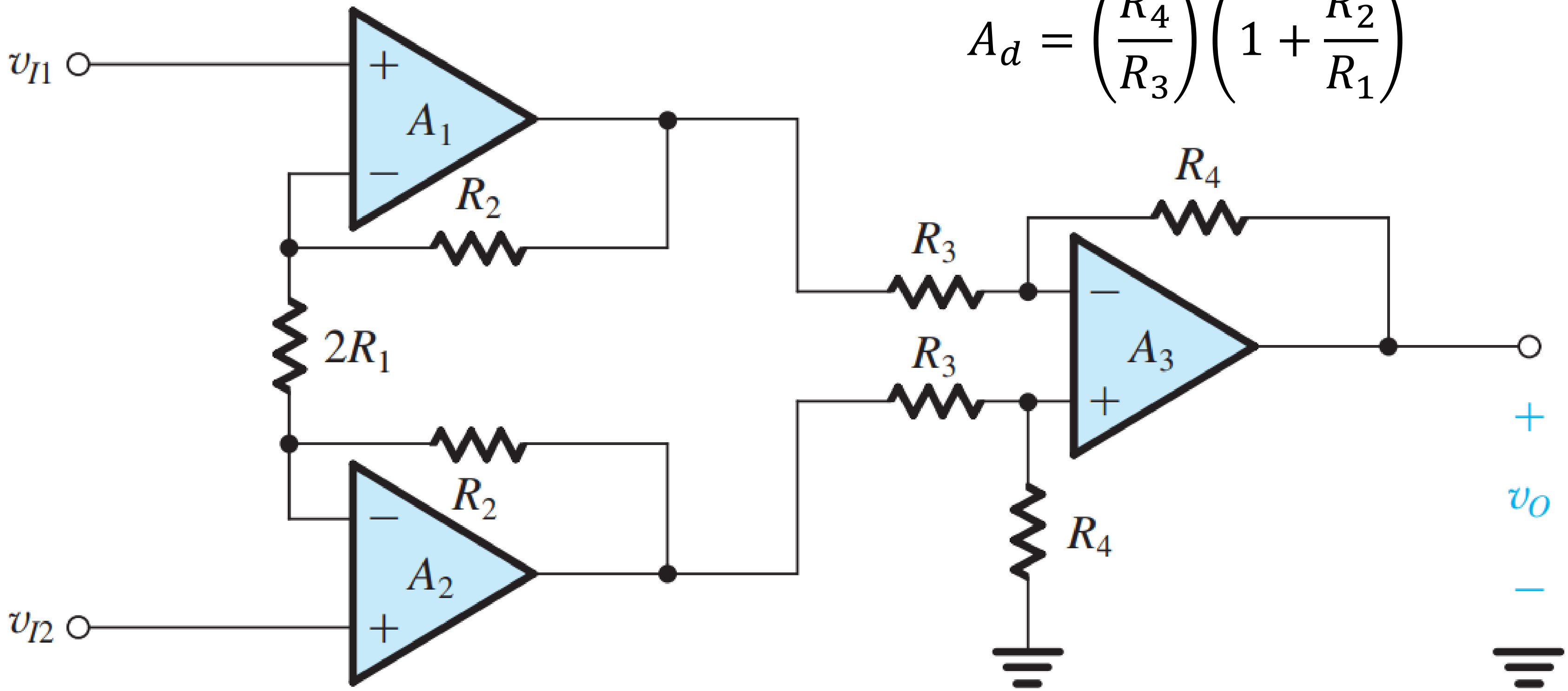
$$v_o = \frac{R_4}{R_3}\left(1+\frac{R_2}{R_1}\right)v_{Id}$$

$$A_d = \left(\frac{R_4}{R_3}\right)\left(1+\frac{R_2}{R_1}\right)$$

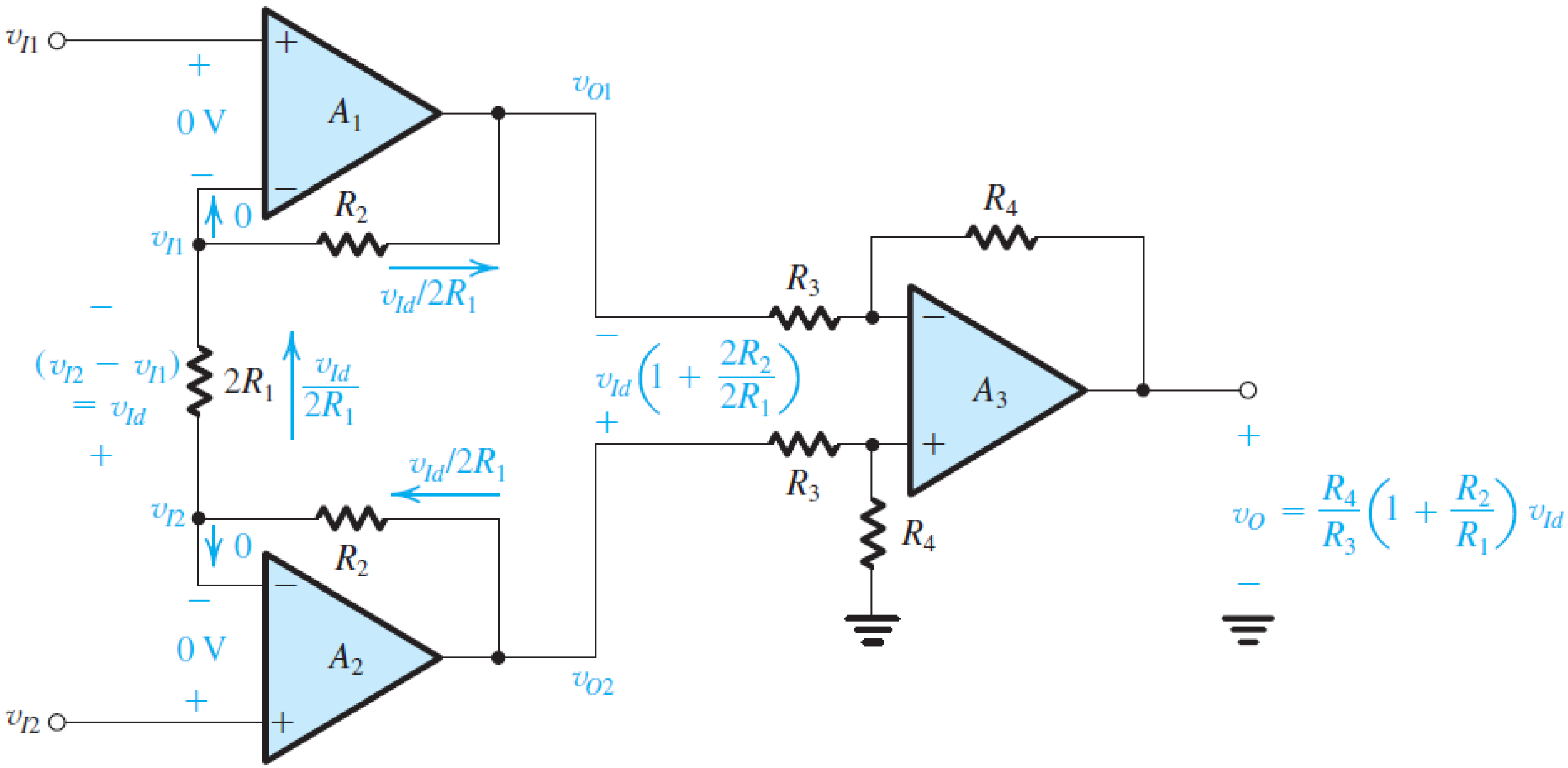
Instrumentation Amplifier

$$v_o = \frac{R_4}{R_3}\left(1 + \frac{R_2}{R_1}\right) v_{Id}$$

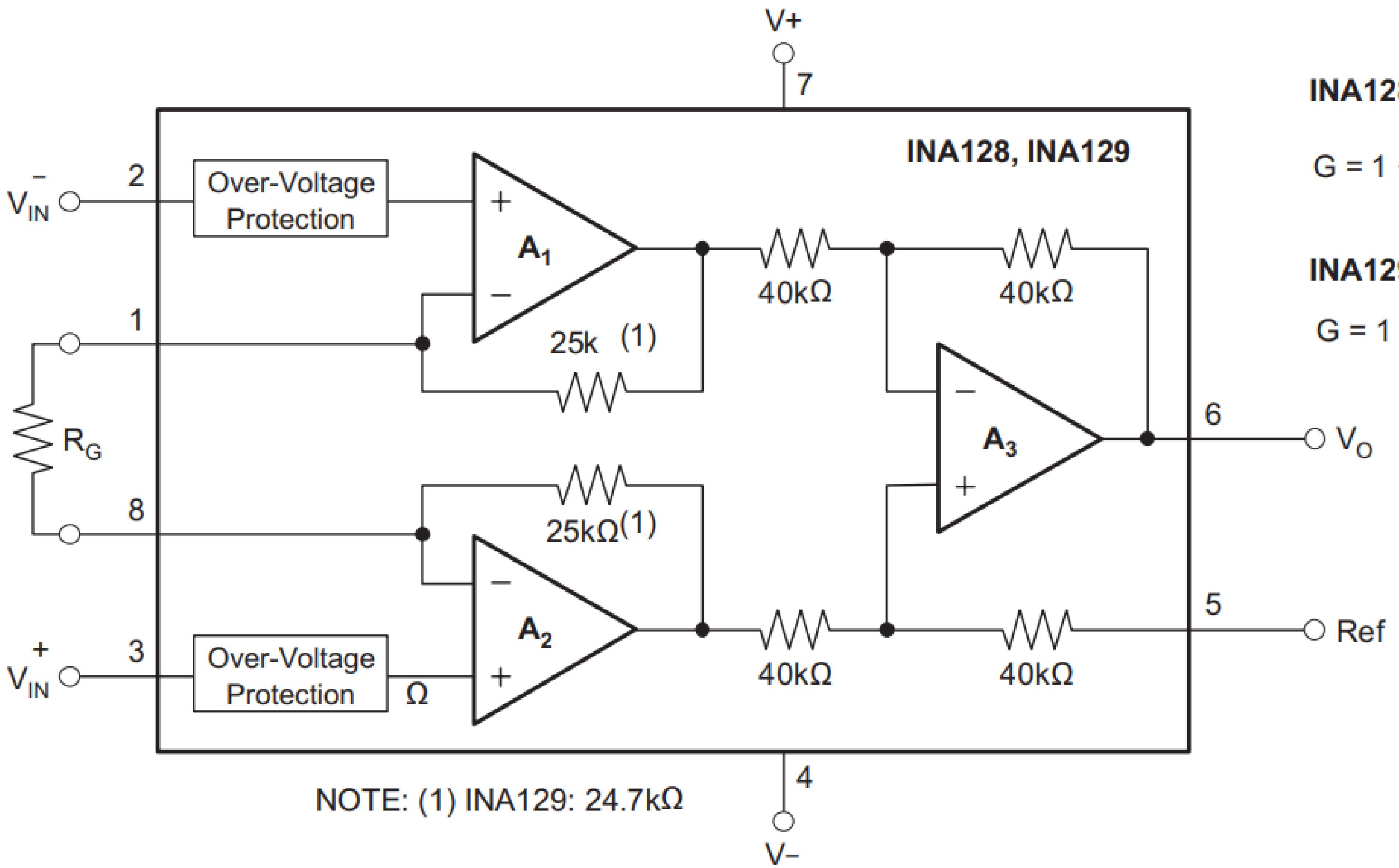
$$A_d = \left(\frac{R_4}{R_3}\right)\left(1 + \frac{R_2}{R_1}\right)$$



Instrumentation Amplifier



Instrumentation Amplifier



INA128:

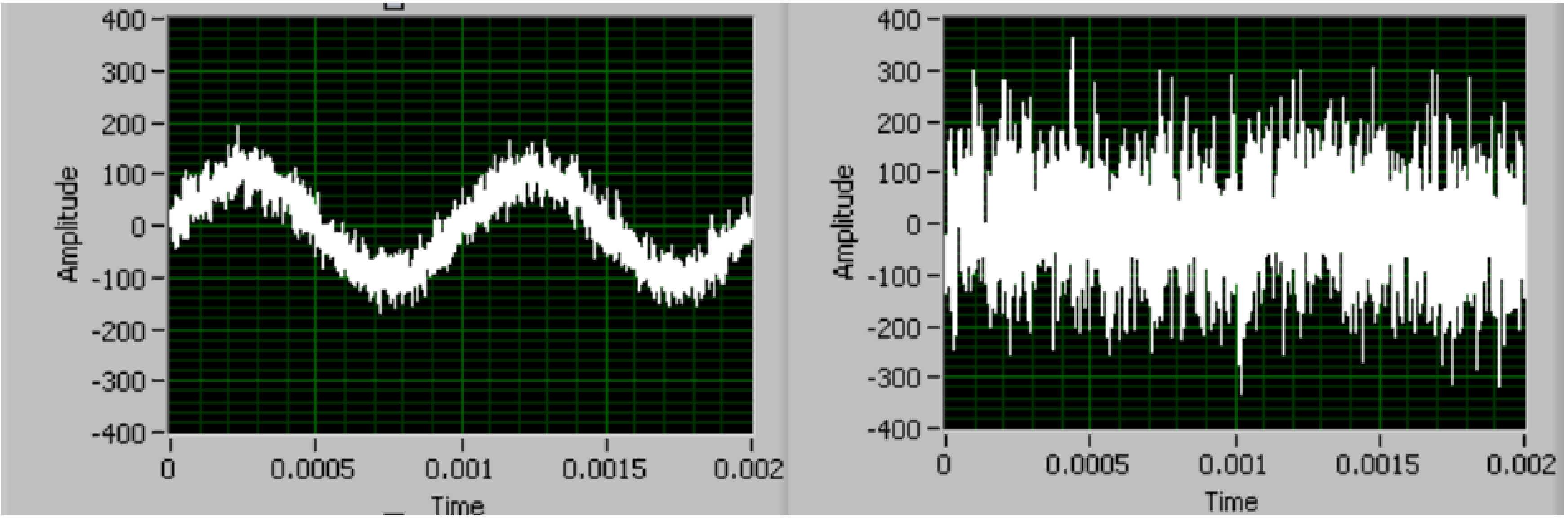
$$G = 1 + \frac{50k\Omega}{R_G}$$

INA129:

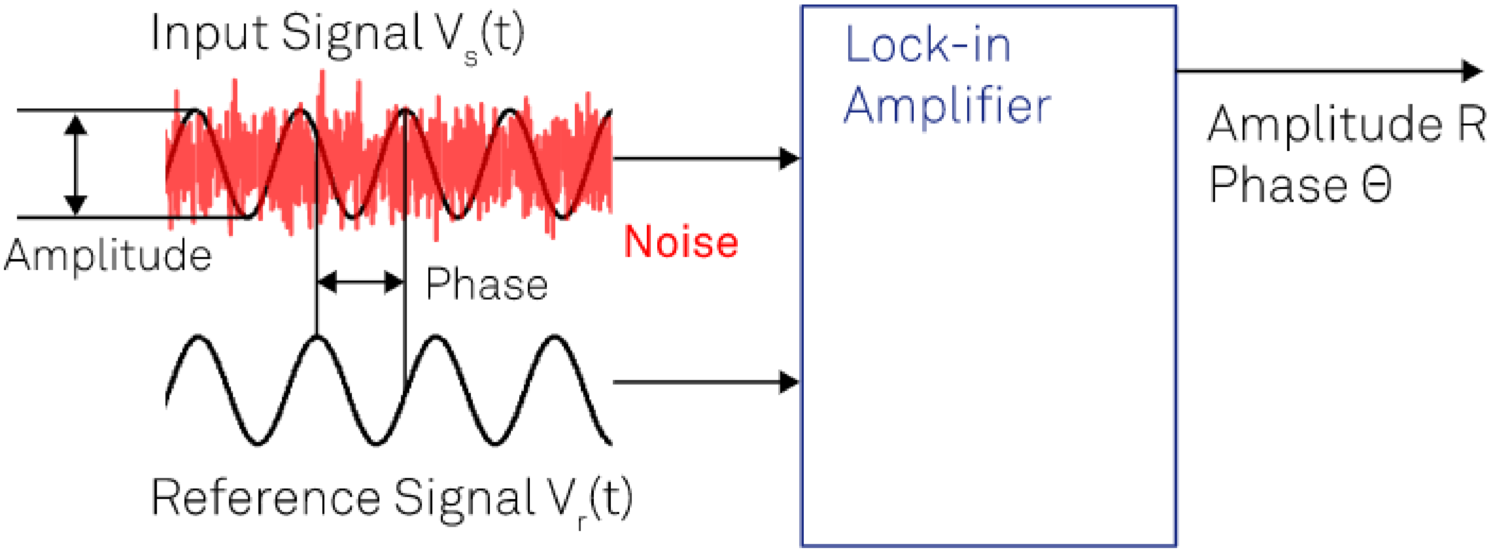
$$G = 1 + \frac{49.4k\Omega}{R_G}$$

Lock-in amplifier

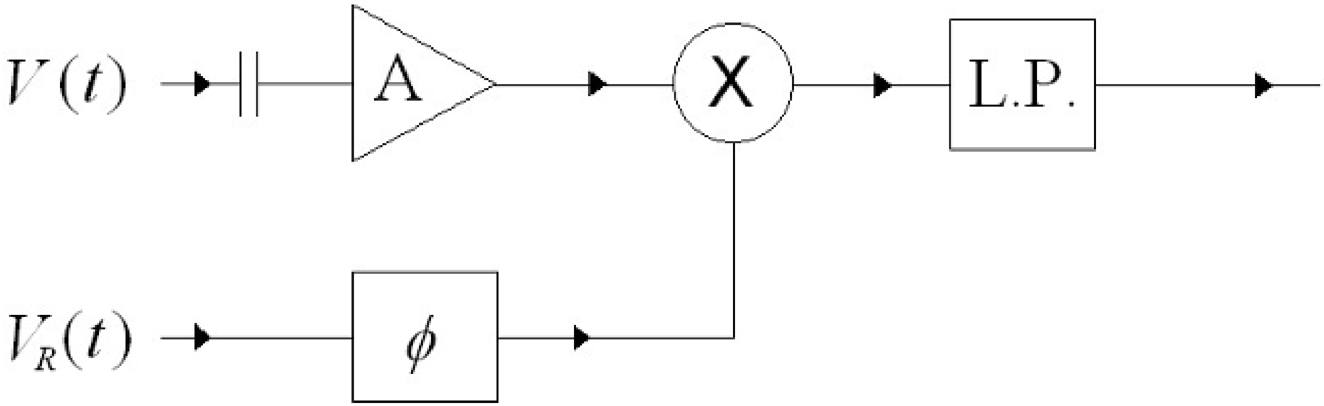
- Detect and measure very small AC signals (down to a few nanovolts) even when the small signal is obscured by larger noise source.



Lock-in Amplifier



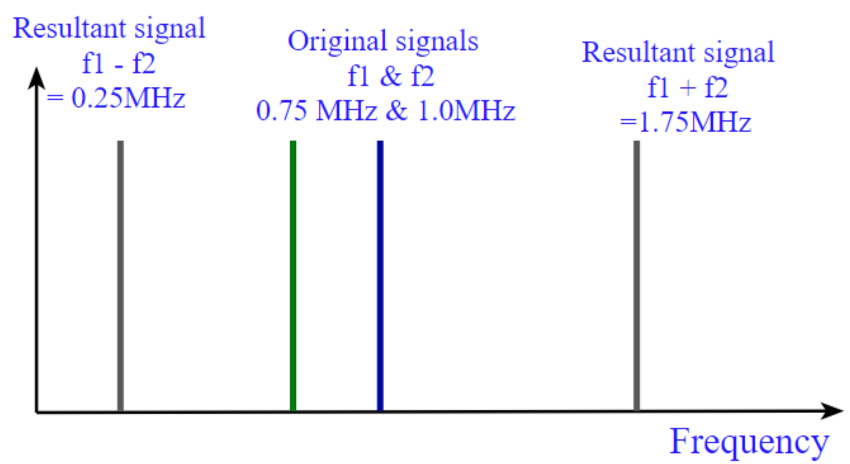
Lock-in amplifier



Lock-in amplifier

- Signal of interest: $V_{sig}\sin(\omega_r t + \theta_{sig})$
- Reference signal: $V_L\sin(\omega_o t + \theta_{ref})$

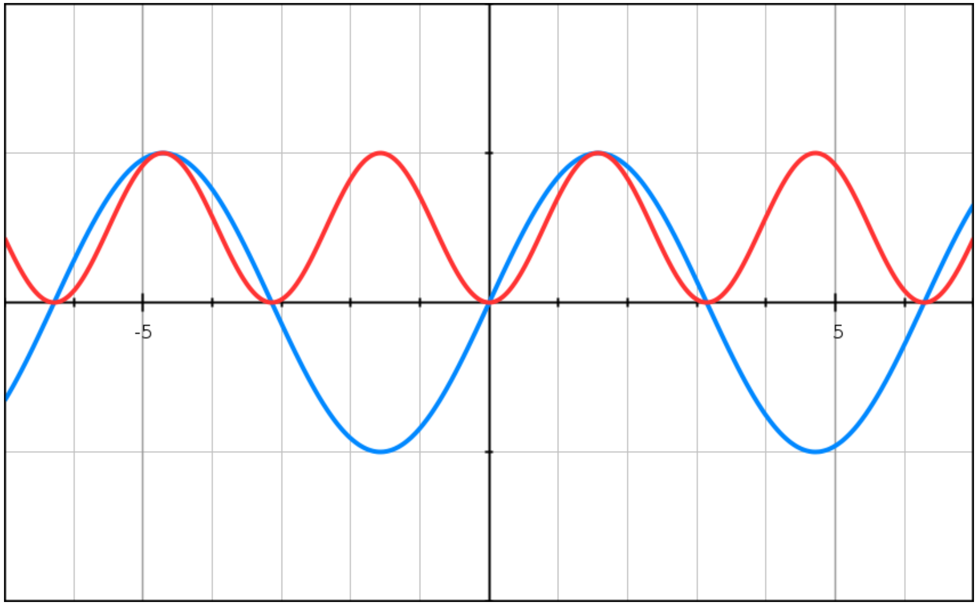
$$V_{psd} = V_{sig}V_L\sin\left(\omega_r t + \theta_{sig}\right)\sin\left(\omega_L t + \theta_{ref}\right)$$
$$= \frac{1}{2}V_{sig}V_L\cos\left([\omega_r - \omega_L]t + \theta_{sig} - \theta_{ref}\right)$$
$$- \frac{1}{2}V_{sig}V_L\cos\left([\omega_r + \omega_L]t + \theta_{sig} + \theta_{ref}\right)$$

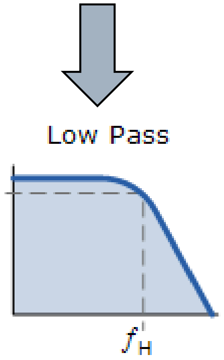


Lock-in amplifier

- If $\omega_L = \omega_r$, the difference frequency component will be a DC signal.

$$V_{psd} = \frac{1}{2}V_{sig}V_L\cos(\theta_{sig} - \theta_{ref}) - \underline{\frac{1}{2}V_{sig}V_L\cos(2\omega t + \theta_{sig} + \theta_{ref})}$$

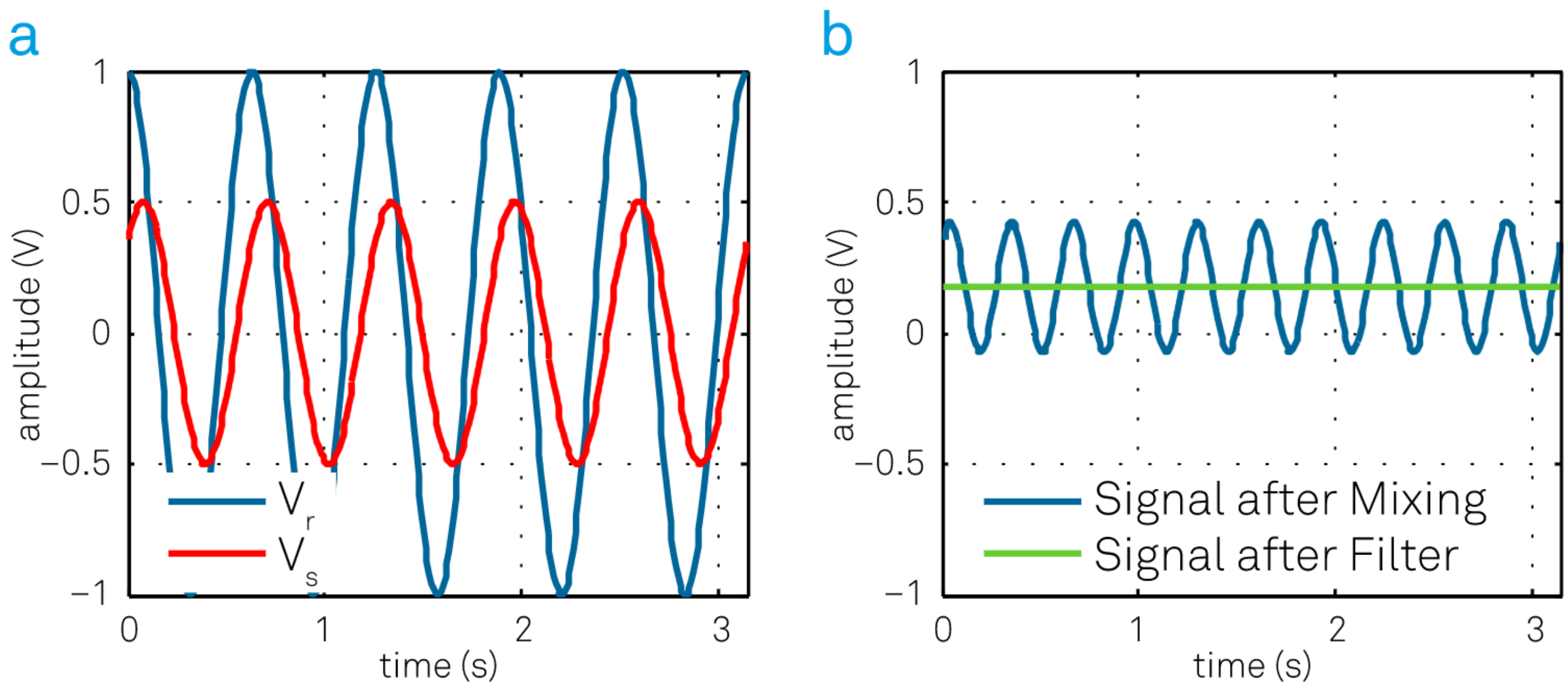




Lock-in amplifier

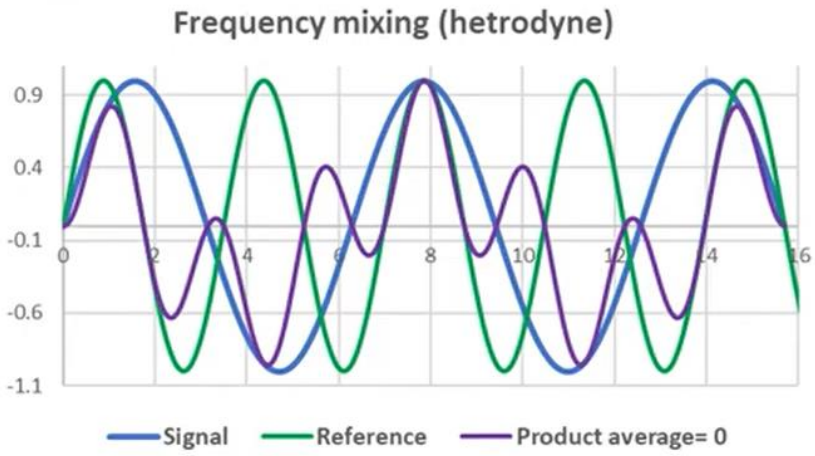
- If $\omega_L = \omega_r$, the difference frequency component will be a DC signal.

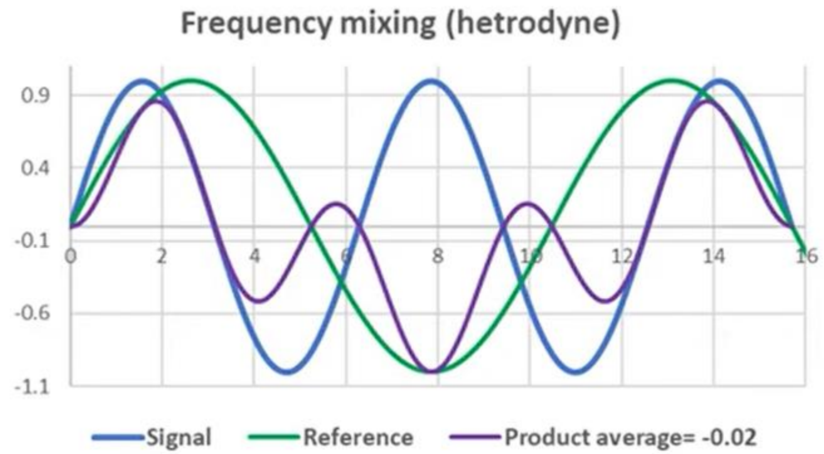
$$V_{psd} = \frac{1}{2} V_{sig} V_L \cos(\theta_{sig} - \theta_{ref}) - \frac{1}{2} V_{sig} V_L \cos(2\omega t + \theta_{sig} + \theta_{ref})$$

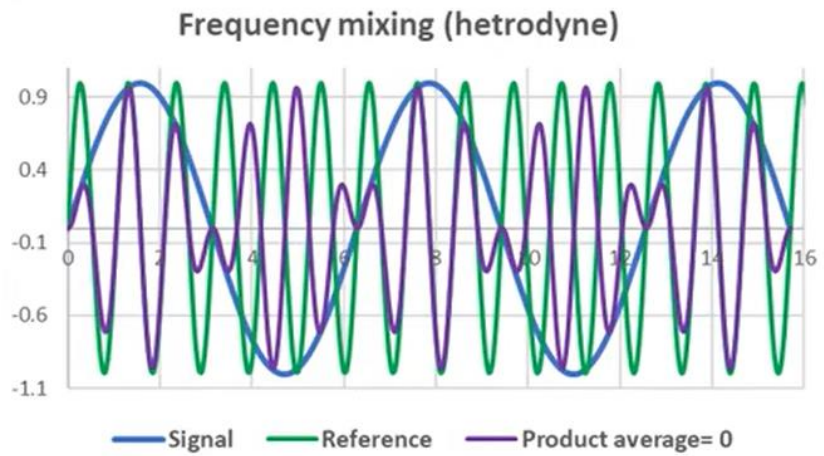


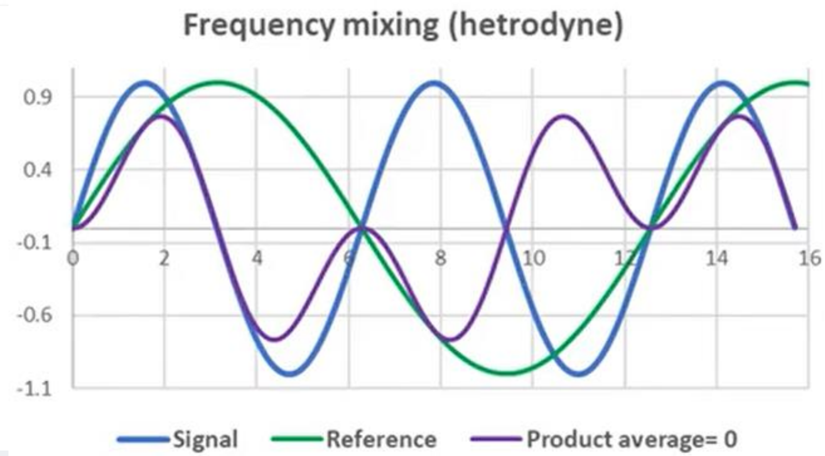
Lock-in amplifier

- If $\omega_L \neq \omega_r$, signal components with different frequencies make new signals, but they still have **an average vale of zero**





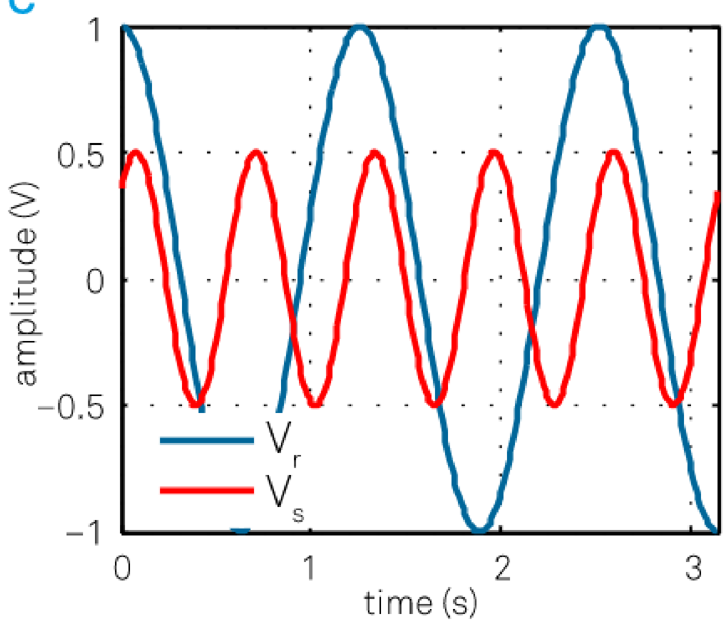




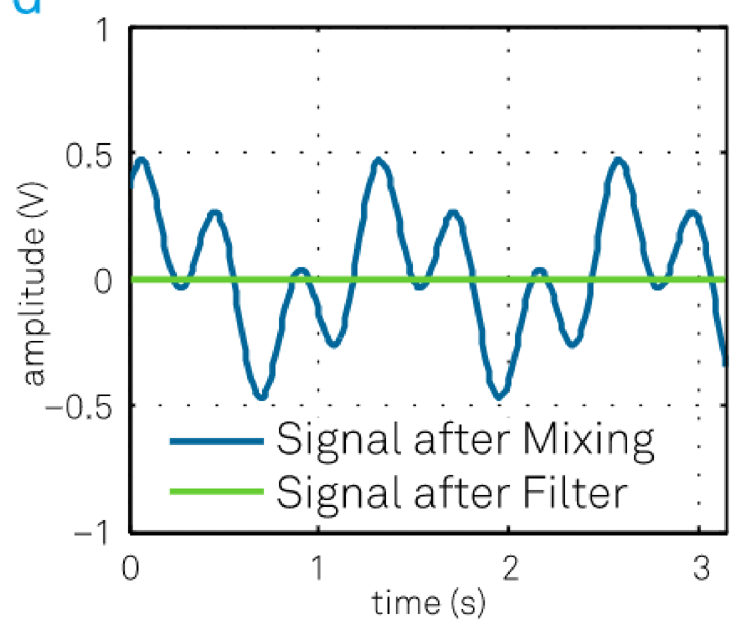
Lock-in amplifier

- If $\omega_L \neq \omega_r$, signal components with different frequencies make new signals, but they still have **an average vale of zero**

c

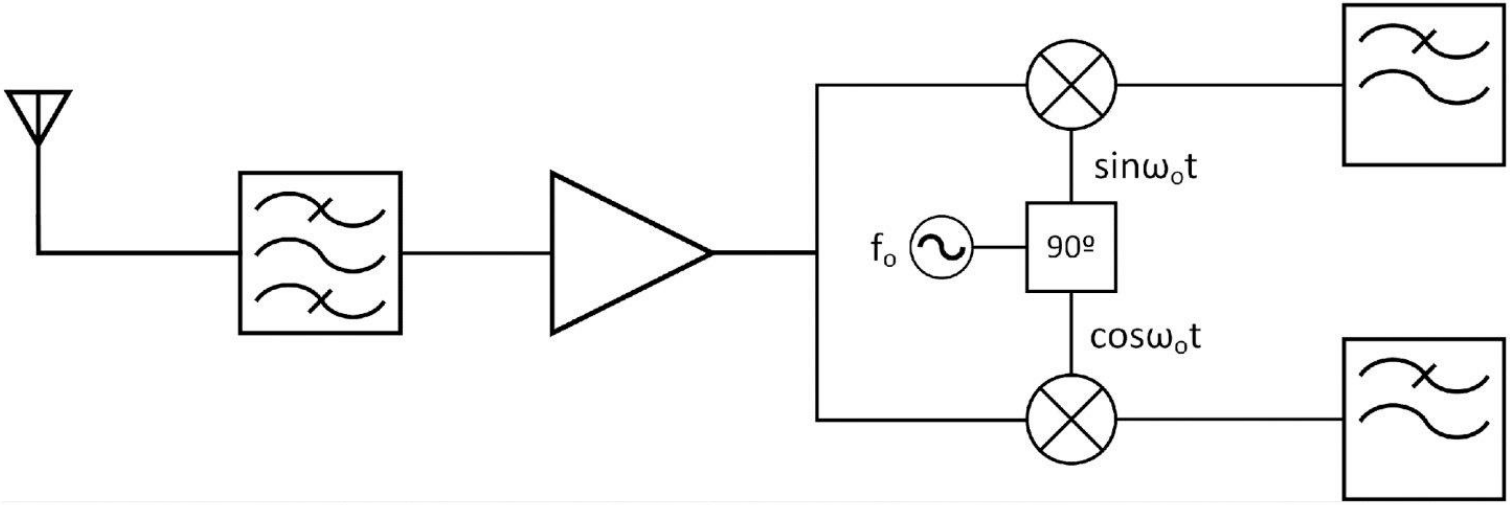


d



Lock-in amplifier

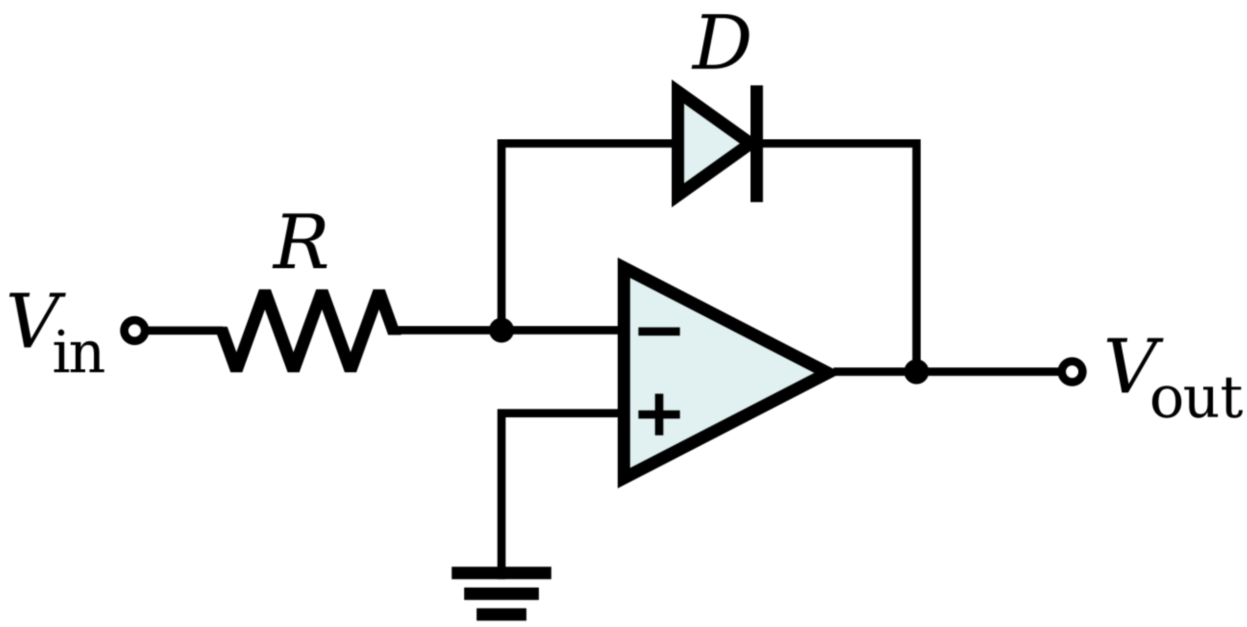
- $X = \frac{1}{2} V_{sig} V_L \cos(\theta_{sig} - \theta_{ref})$
- $Y = \frac{1}{2} V_{sig} V_L \sin(\theta_{sig} - \theta_{ref})$



$$A = 2 \times \sqrt{X^2 + Y^2}$$

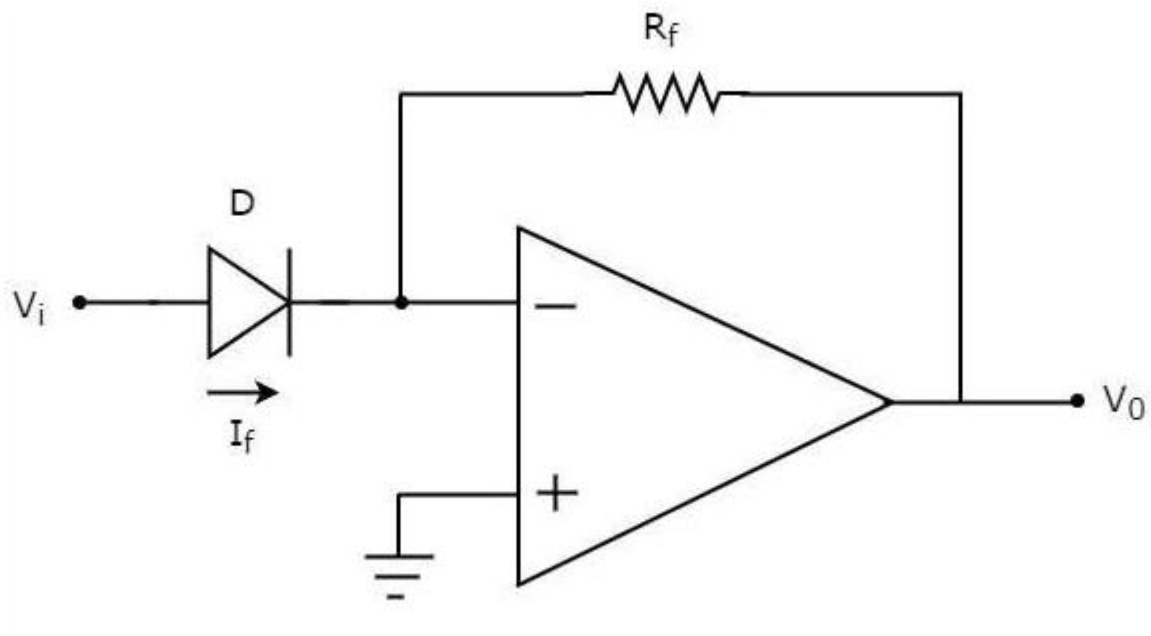
$$Phase = \tan^{-1} \frac{Y}{X}$$

Log Amplifier



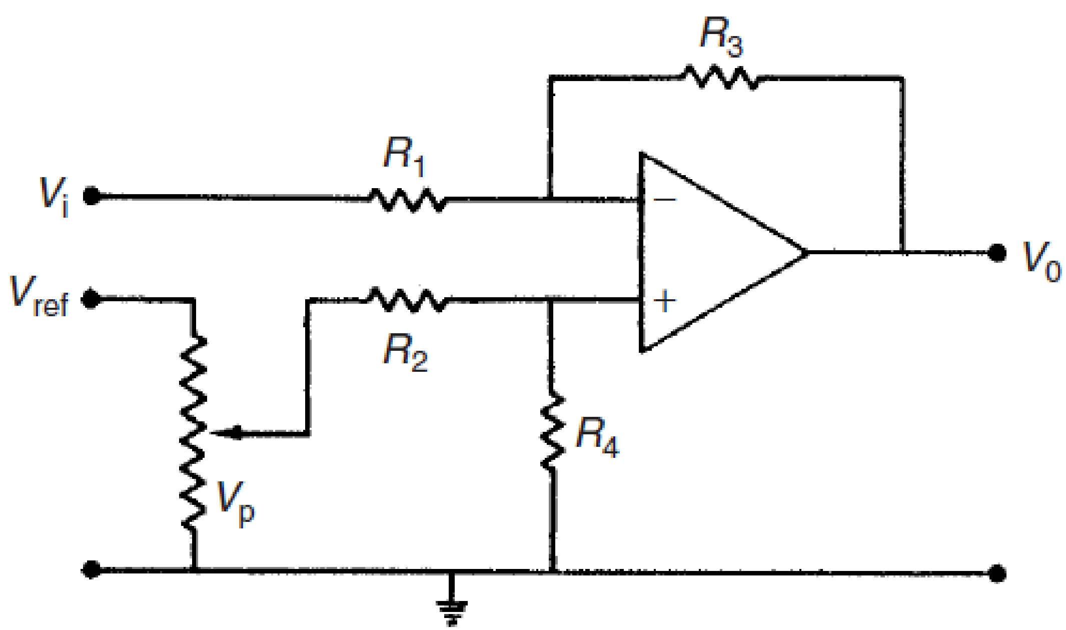
$$V_{out} = -V_T \ln \frac{V_{in}}{I_s R}$$

Anti-Log Amplifier

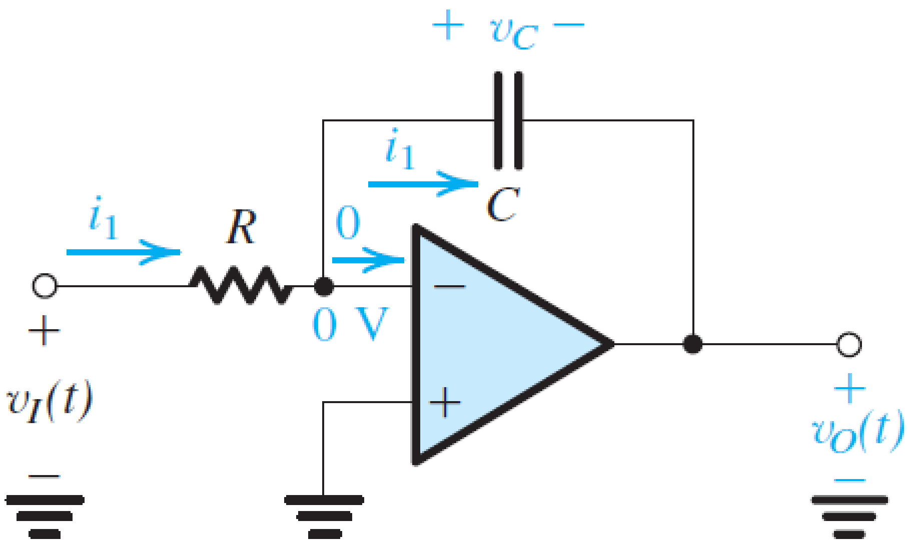


$$V_{out} = -R_f I_s e^{\left(\frac{V_i}{V_T}\right)}$$

Bias (zero drift) removal



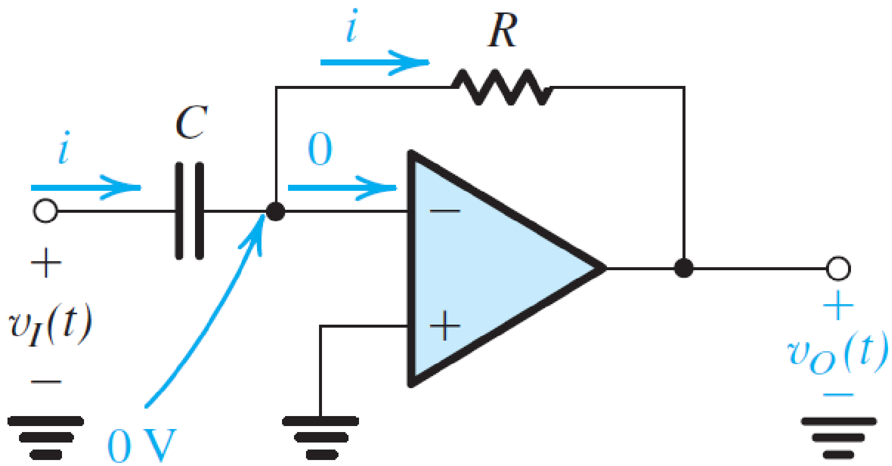
Signal integration



$$v_O(t) = -\frac{1}{CR}\int_0^t v_I(t)\,dt - V_C$$

$$\frac{V_o}{V_i} = -\frac{1}{sCR}$$

Signal differentiation



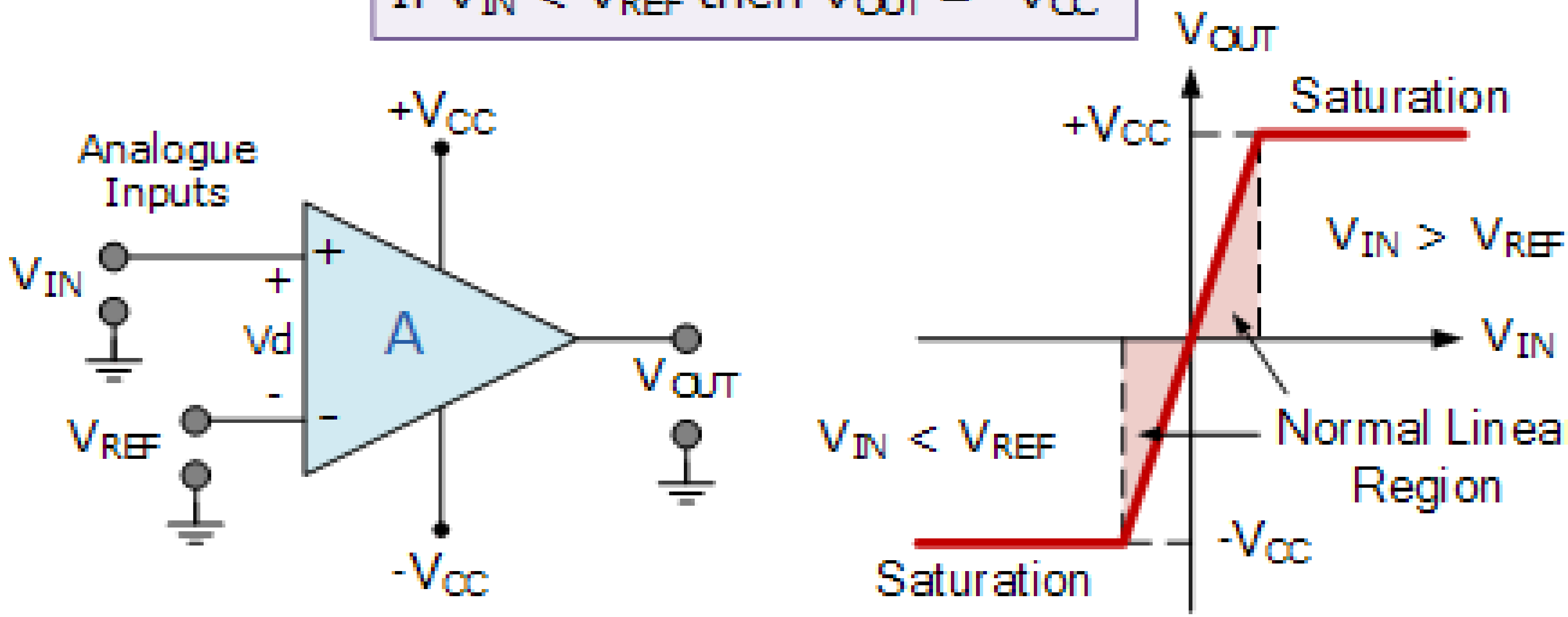
$$i(t) = C\frac{dv_I(t)}{dt}$$

$$v_O(t) = -CR\frac{dv_I(t)}{dt}$$

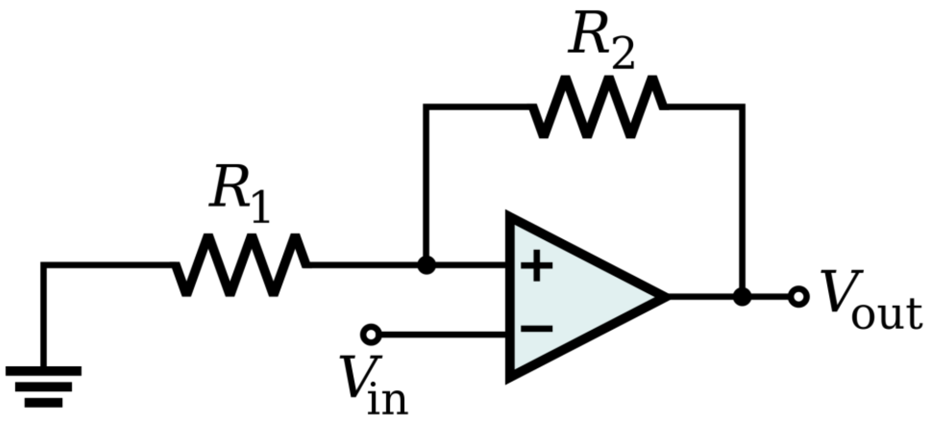
$$\frac{V_o}{V_i} = -sCR$$

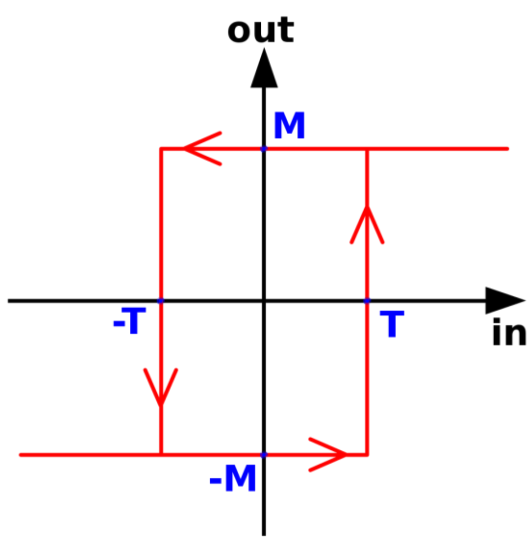
Voltage comparator

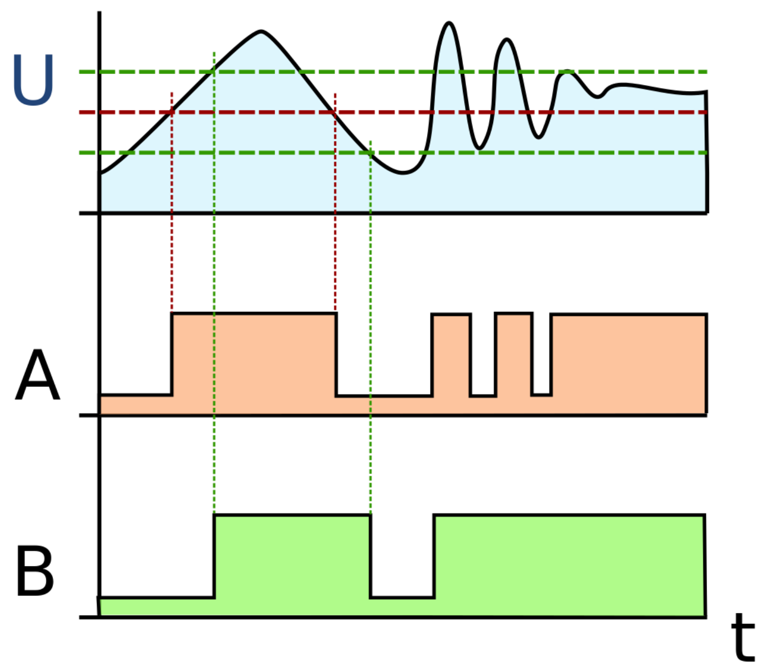
If $V_{IN} > V_{REF}$ then $V_{OUT} = +V_{CC}$
If $V_{IN} < V_{REF}$ then $V_{OUT} = -V_{CC}$



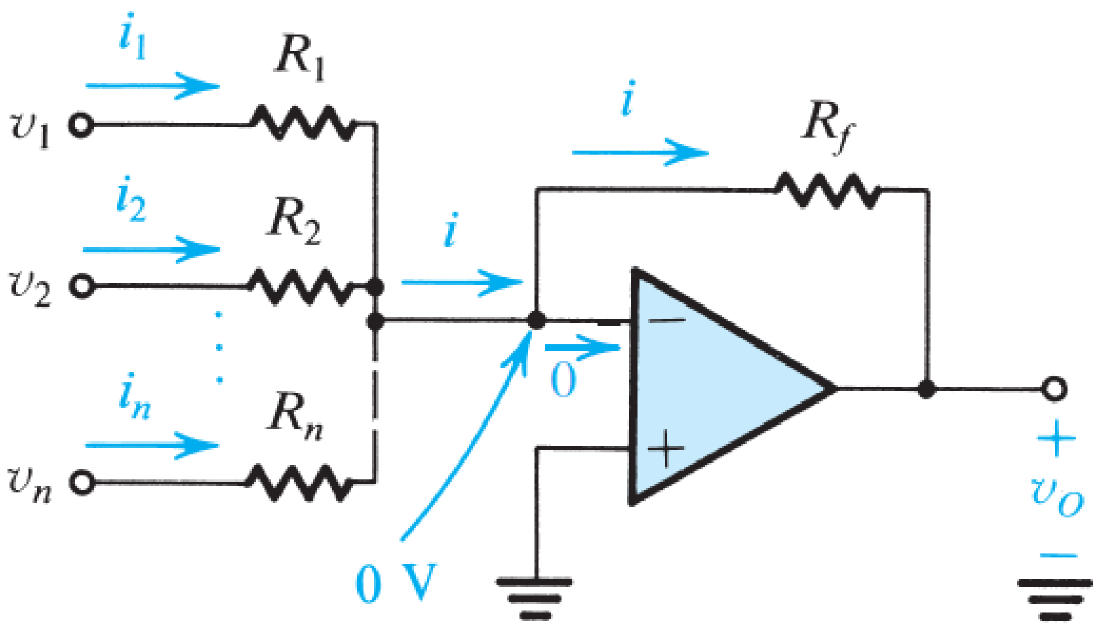
Schmitt trigger





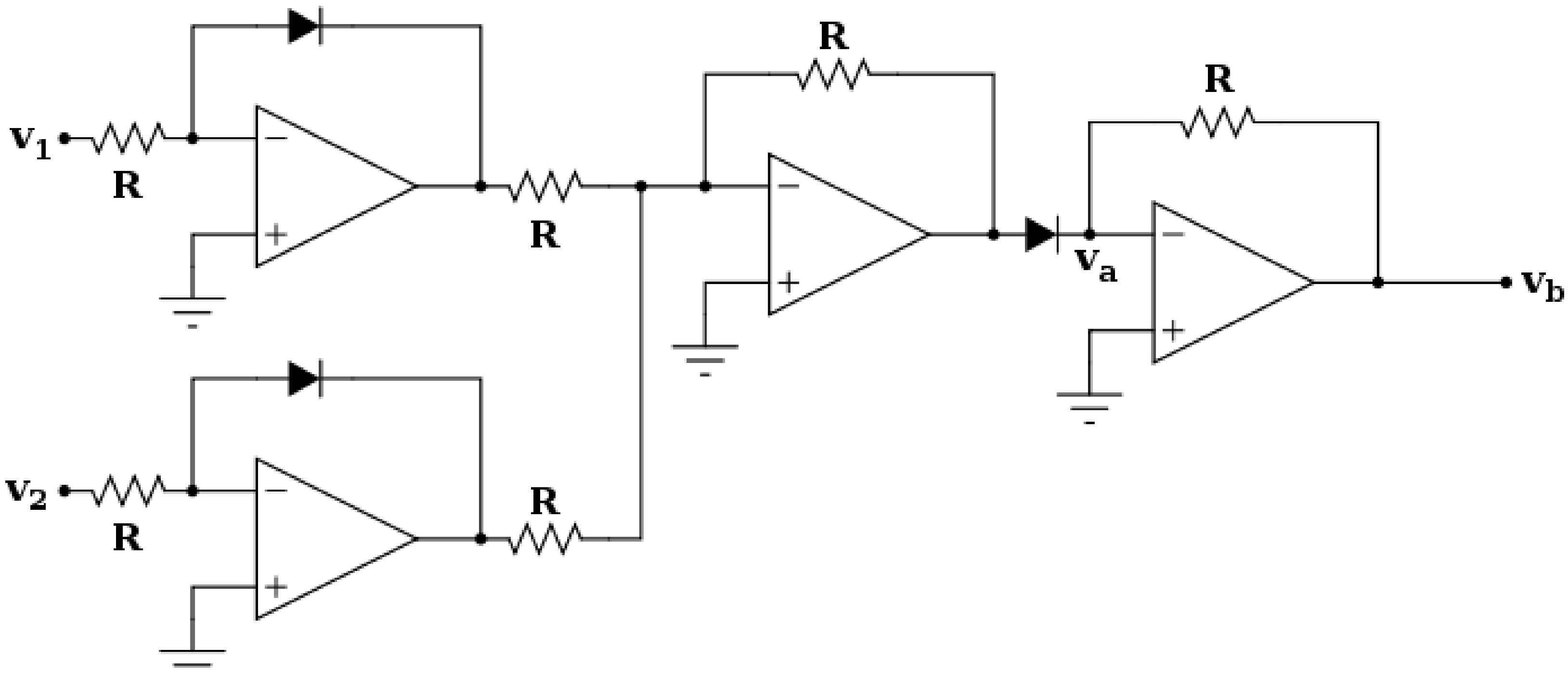


Signal addition

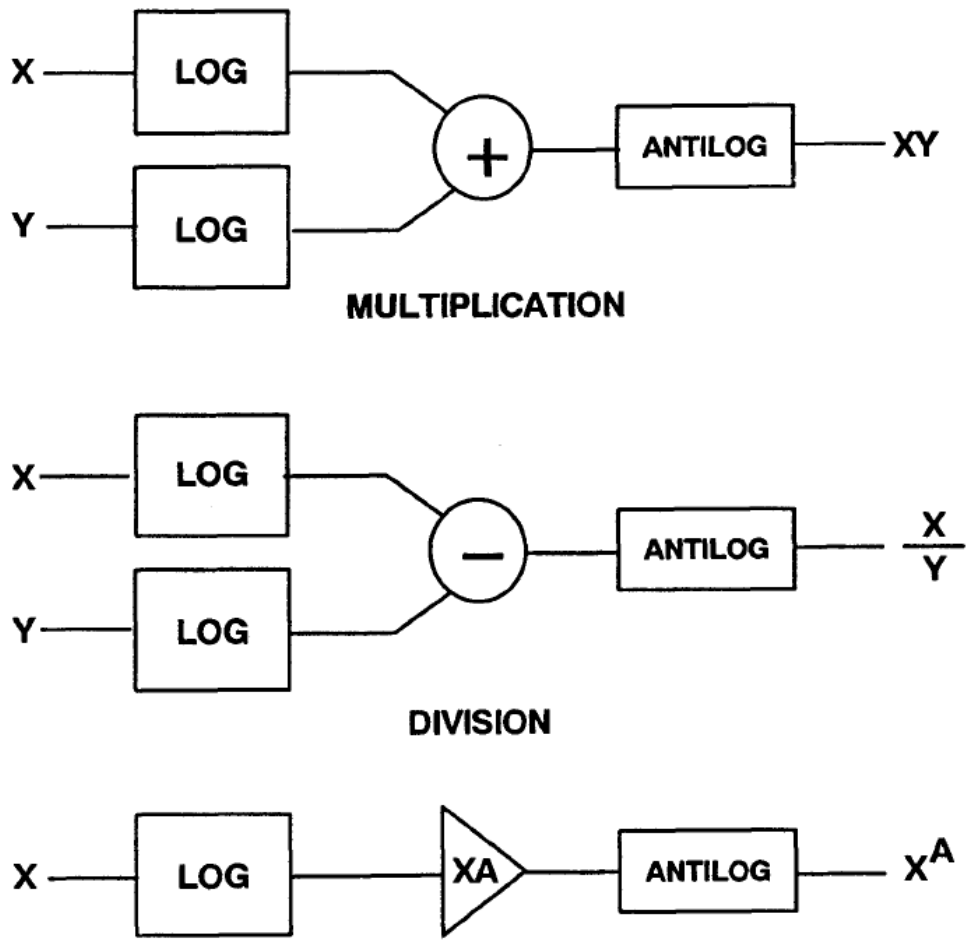


$$v_O = -\left(\frac{R_f}{R_1}v_1 + \frac{R_f}{R_2}v_2 + \cdots + \frac{R_f}{R_n}v_n\right)$$

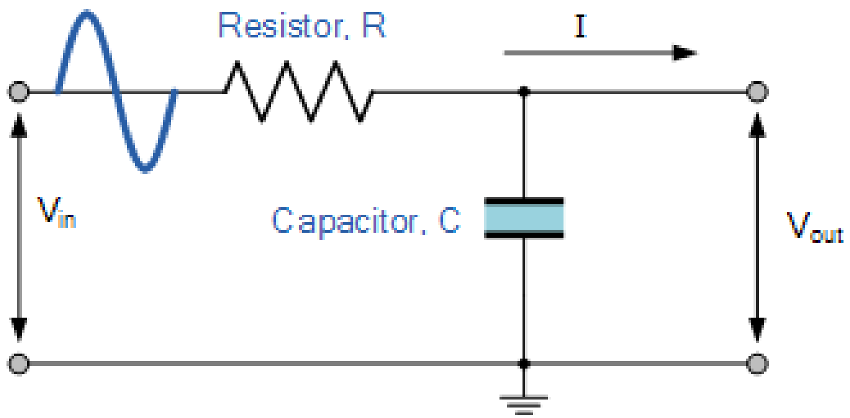
Signal multiplication



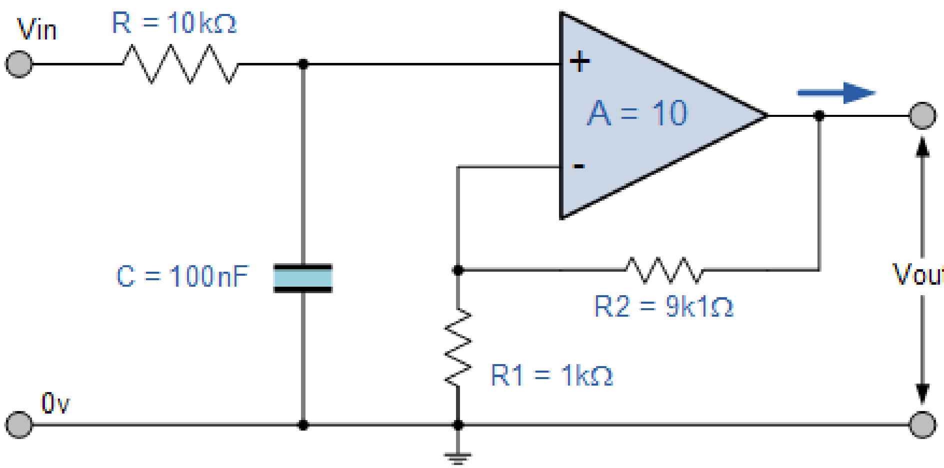
Signal multiplication



Low Pass Filter

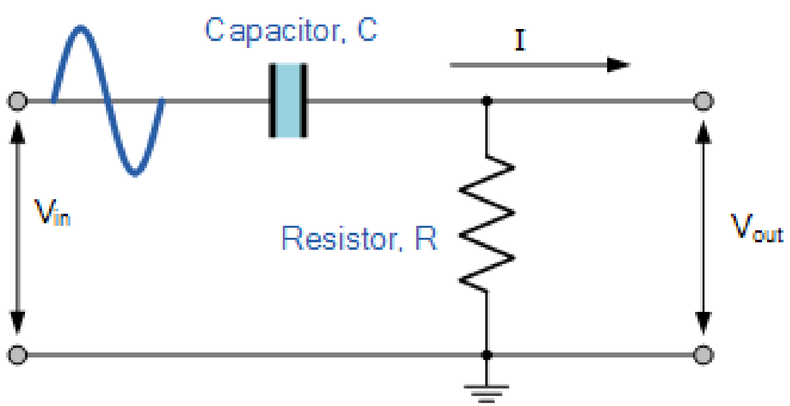


Passive Low Pass Filter

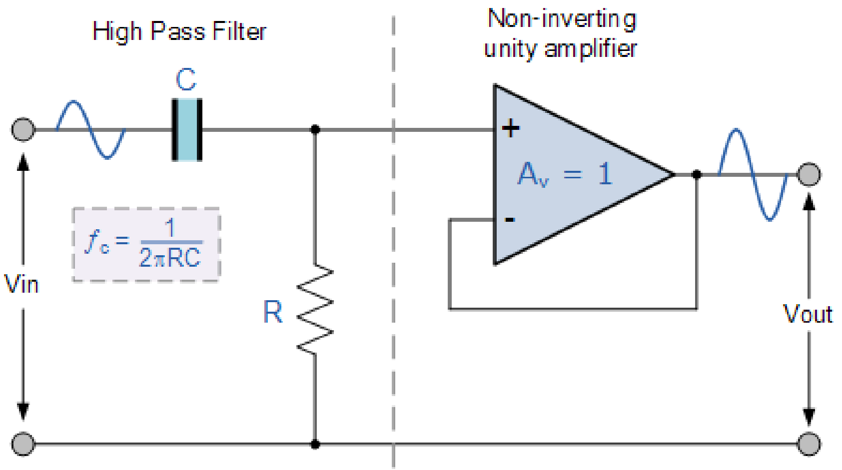


Active Low Pass Filter

High Pass Filter



Passive High Pass Filter



Active High Pass Filter